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Machine Learning Lab

Quality Laboratory Manual

Experiment No. 7

Study and Implementation of K-Means Clustering Algorithm

Course Instructor -

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Experiment No. 7

Title of Experiment: To study and implement K-Means Clustering Algorithm

Aim of Experiment: To implement and understand the working of K-Means clustering algorithm, an unsupervised learning technique used for grouping data point into cluster based on their similarity in Machine Learning

System Requirements – Win 8 and above OS, 4GB RAM, 2.33 GHz Processor

Software/s Needed for Experiment – Jupyter Notebook/ Anaconda Navigator/ Google Colaboratory/ Spyder, Python 3.x [With libraries such as Numpy, Pandas, Matplotlib and Scikit-Learn]

Experiment Outcomes –

1. Understand the concept and purpose of K-Means clustering algorithm in unsupervised learning
2. To understand and implement how K-Means groups data points into clusters based on similarity
3. To understand the impact of the number of clusters ‘k’ on the clustering results
4. To create scatter plots to visualize clusters and their centroids

Theory –

K-means clustering is a popular unsupervised machine learning algorithm used for partitioning a dataset into a pre-defined number of clusters. The goal is to group similar data points together and discover underlying patterns or structures within the data.

Recall the first property of clusters – it states that the points within a cluster should be similar to each other. So, our aim here is to minimize the distance between the points within a cluster. There is an algorithm that tries to minimize the distance of the points in a cluster with their centroid – the k-means clustering technique.

K-means is a centroid-based algorithm or a distance-based algorithm, where we calculate the distances to assign a point to a cluster. In K-Means, each cluster is associated with a centroid.

The main objective of the K-Means algorithm is to minimize the sum of distances between the points and their respective cluster centroid. Optimization plays a crucial role in the k-means clustering algorithm. The goal of the optimization process is to find the best set of centroids that minimizes the sum of squared distances between each data point and its closest centroid

How K-Means clustering works?

1) **Initialization:** Start by randomly selecting K points from the dataset. These points will act as the initial cluster centroids.

2) **Assignment:** For each data point in the dataset, calculate the distance between that point and each of the K centroids. Assign the data point to the cluster whose centroid is closest to it. This step effectively forms K clusters.

3) **Update centroids:** Once all data points have been assigned to clusters, recalculate the centroids of the clusters by taking the mean of all data points assigned to each cluster.

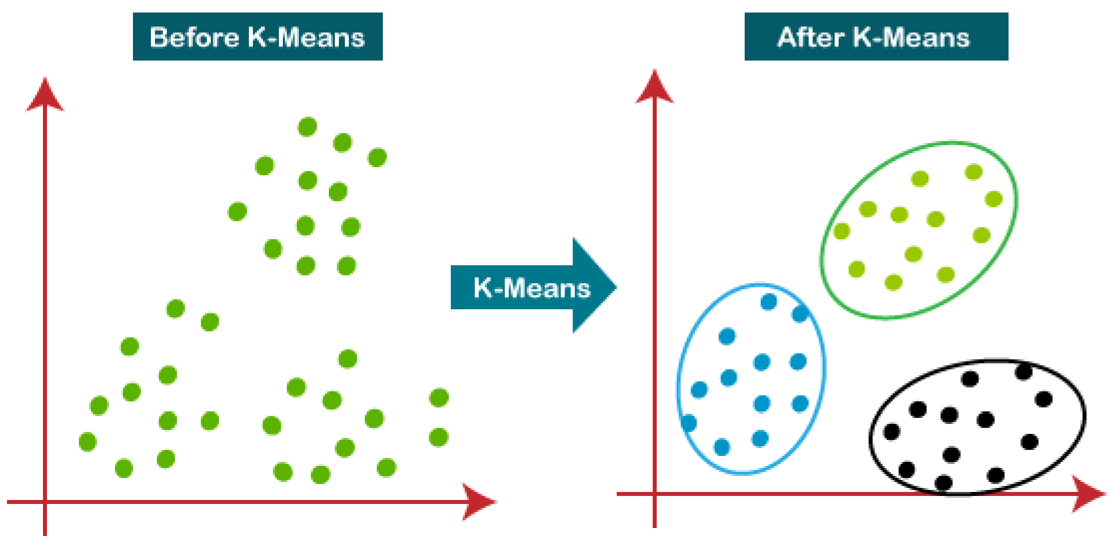
4) **Repeat:** Repeat steps 2 and 3 until convergence. Convergence occurs when the centroids no longer change significantly or when a specified number of iterations is reached.

5) **Final Result:** Once convergence is achieved, the algorithm outputs the final cluster centroids and the assignment of each data point to a cluster

The algorithm takes the unlabeled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters. The value of k should be predetermined in this algorithm.

The k-means clustering algorithm mainly performs two tasks:

- Determines the best value for K center points or centroids by an iterative process
- Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster



Procedure to implement K-Means clustering algorithm:

1. **Data Collection:** Obtain or create a dataset suitable for clustering. A dataset such as Iris, which classifies iris flowers into three species – Setosa, Versicolour and Virginica

2. **Data Preprocessing:**
    a. Load the dataset into a Pandas DataFrame.
    b. Standardize the features if necessary, especially if the features have different scales

3. **Implementing K-Means Clustering:** Use the K-Means class from scikit learn to implement the K-Means algorithm. Hence choose the number of clusters 'k' and train the model of the dataset

4. **Cluster Assignment and Centroid Calculations:** Assign each data point to the nearest cluster based on the Euclidean distance and then extract the cluster centroids

5. **Visualize the clusters:**
    a. Visualize the clusters and their centroids using scatter plots
    b. Use the Elbow method to determine the optimal value of 'k'

6. **Model Evaluation:**
    a. Evaluate the clustering using metrics
    b. Analyze the results and discuss the insights obtained from the clustering

Sample Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

# Step 1: Data Collection (Iris dataset)
iris = load_iris()
X = pd.DataFrame(iris.data, columns=iris.feature_names)

# Step 2: Data Preprocessing
# Standardizing the data (optional but recommended)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Step 3: Implementing K-Means Clustering
# Choosing number of clusters (K)
kmeans = KMeans(n_clusters=3, random_state=42)
```

```
kmeans.fit(X_scaled)

# Step 4: Cluster Assignment and Centroid Calculation
X['cluster'] = kmeans.labels_  # Assign cluster labels to data
points
centroids = kmeans.cluster_centers_

# Step 5: Visualizing Clusters
plt.figure(figsize=(8, 6))
sns.scatterplot(x=X_scaled[:, 0], y=X_scaled[:, 1],
hue=X['cluster'], palette='Set1', legend='full')
plt.scatter(centroids[:, 0], centroids[:, 1], s=100, color='black',
marker='X')  # Plot centroids
plt.title('K-Means Clustering of Iris Dataset')
plt.xlabel('Feature 1 (Standardized)')
plt.ylabel('Feature 2 (Standardized)')
plt.show()

# Step 6: Evaluating the Model using Inertia and Silhouette Score
inertia = kmeans.inertia_
silhouette_avg = silhouette_score(X_scaled, kmeans.labels_)

print(f"Inertia: {inertia}")
print(f"Silhouette Score: {silhouette_avg}")

# Step 7: Using the Elbow Method to Find Optimal K
inertias = []
K = range(1, 11)
for k in K:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertias.append(kmeans.inertia_)

plt.figure(figsize=(8, 6))
plt.plot(K, inertias, marker='o')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Inertia')
plt.show()
```

Observations -

- Record the assigned clusters for each data point
- Observe the distribution of data points across different clusters

Conclusion –

Hence, the model summarizes the findings from the experiment, such as the quality of the clusters and the significance of feature standardization. Also the importance of the right number of clusters and how K-Means can be applied to solve real-world problems

References –

a. **Textbook –**
   i. Machine Learning with Python – An approach to Applied ML – Abhishek Vijayvargiya, BPB Publications, 1st Edition 2018
   ii. Machine Learning, Tom Mitchell, McGraw Hill Education, 1st Edition 1997

b. **Online references –**
   i. https://www.analyticsvidhya.com/blog/2019/08/comprehensive-guide-k-means-clustering/
   ii. https://medium.com/@sayahfares19/k-means-clustering-algorithm-for-unsupervised-learning-tasks-f761ed7f37c0
   iii. https://towardsdatascience.com/understanding-k-means-clustering-in-machine-learning-6a6e67336aa1

Expected Oral Questions –

1. What is a K-Means clustering and how does it work?
2. Is K-Means a supervised or unsupervised learning algorithm? Why?
3. What type of problems can be solved using K-Means clustering?
4. What are the components of K-Means clustering?
5. What is the objective function of K-Means? How is it minimized?
6. How does the K-Means algorithm initialize the cluster centroids?
7. Explain the iterative process used in K-Means clustering.
8. What are some methods to choose the optimal value of ‘k’?
9. What is elbow method and how is it applied in K-Means?
10. How do you interpret the final clusters in a K-Means model?

FAQ’s in Interview –

1. What is K-Means clustering and why is it used?
2. What are centroids in the context of K-Means?
3. Describe the algorithmic steps of the K-Means clustering methods
4. What is the role of distance metrics in K-Means and which distances can be used?
5. How do you decide on the number of clusters ‘k’ in a K-Means algorithm?